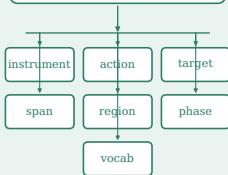


1. Data Enrichment

heterogeneous sources

8 corpora, 6,270 rows



7 typed streams $\rightarrow$ auxiliary QA

8,150

6,270

2. Training

gradient share per (dataset, task)



uniform vs $\tau = 0.5$

Qwen3-VL-4B

LoRA adapters



corrupted-prior conditioning

3. Self-Refining Inference

temporal zoom



resample inside hypothesis

spatial zoom



coarse box

crop, re-predict

4. Ordinal Decoding

$K = 5$ sampled generations

0 1 1 0 2

$\hat{v} = \text{round}(\mathbb{E}[v])$

0.8 $\rightarrow$ 1

recovers intermediate scores

greedy

returns 0

5. Output Calibration

budget allocation
from reference length

question-aware keying
(id, task, question)

span and box completion
clamp, replicate

terminology normalisation
ontology lookup

benchmark-compliant output